

Customer Shopping Behavior Analysis

An end-to-end analysis of retail transaction data — Python (EDA) · PostgreSQL (SQL analysis) · Power BI (dashboard)

1. Project Overview

This project examines customer shopping behavior using a transactional dataset comprising 3,900 purchase records across multiple product categories. The objective is to surface actionable insights into customer spending patterns, segmentation, product performance, and subscription behavior in order to support data-driven business decisions.

2. Dataset Summary

The dataset consists of 3,900 rows and 18 columns, spanning customer demographics, purchase details, and shopping behavior indicators.

Attribute	Details
Rows	3,900
Columns	18
Customer demographics	Age, Gender, Location, Subscription Status
Purchase details	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
Missing data	37 missing values in the Review Rating column

3. Exploratory Data Analysis (Python)

Data preparation and cleaning were carried out in Python prior to analysis:

- Data loading** — the dataset was imported using pandas.
- Initial exploration** — `df.info()` and `df.describe()` were used to review structure and summary statistics.
- Missing data handling** — null values were identified, and missing entries in the Review Rating column were imputed using the median rating within each product category.
- Column standardization** — column names were converted to `snake_case` to improve readability and documentation consistency.

- **Feature engineering** — an `age_group` column was derived by binning customer ages, and a `purchase_frequency_days` column was created from purchase-cycle data.
- **Data consistency check** — the Discount Applied and Promo Code Used fields were found to be redundant; Promo Code Used was subsequently dropped.
- **Database integration** — the cleaned DataFrame was loaded into PostgreSQL to enable structured SQL analysis.

4. Business Analysis (SQL / PostgreSQL)

Structured queries were executed in PostgreSQL to answer ten key business questions.

4.1 Revenue by Gender

Total revenue generated by male versus female customers.

Gender	Revenue (USD)
Female	75,191
Male	157,890

4.2 High-Spending Discount Users

Customers who applied a discount yet still spent above the average purchase amount — 839 customers matched this profile. A representative sample is shown below.

Customer ID	Purchase Amount (USD)
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

839 customers in total met this criterion.

4.3 Top 5 Products by Rating

Products with the highest average customer review ratings.

Item Purchased	Average Rating
Gloves	3.86
Sandals	3.84
Boots	3.82

Item Purchased	Average Rating
Hat	3.80
Skirt	3.78

4.4 Shipping Type Comparison

Average purchase amount by shipping method (Standard vs. Express).

Shipping Type	Avg. Purchase Amount (USD)
Standard	58.46
Express	60.48

4.5 Subscribers vs. Non-Subscribers

Average spend and total revenue by subscription status.

Subscription Status	Customers	Avg. Spend (USD)	Total Revenue (USD)
Yes	1,053	59.49	62,645.00
No	2,847	59.87	170,436.00

4.6 Discount-Dependent Products

Products with the highest share of discounted purchases.

Item Purchased	Discount Rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

4.7 Customer Segmentation

Customers classified into New, Returning, and Loyal segments based on purchase history.

Customer Segment	Number of Customers
Loyal	3,116
Returning	701
New	83

4.8 Top 3 Products per Category

The most frequently purchased products within each product category.

Rank	Category	Item Purchased	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

4.9 Repeat Buyers & Subscriptions

Whether customers with more than five purchases are more likely to subscribe.

Subscription Status	Repeat Buyers
No	2,518
Yes	958

Repeat buyers are not predominantly subscribers, suggesting that purchase frequency alone does not strongly predict subscription uptake.

4.10 Revenue by Age Group

Total revenue contribution by customer age group.

Age Group	Total Revenue (USD)
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

5. Power BI Dashboard

An interactive Power BI dashboard was developed to present the analysis visually, featuring filters for subscription status, gender, category, and shipping type alongside KPI cards (customer count, average purchase amount, average review rating) and breakdowns of revenue and sales by category and by age group.

Metric	Value
Number of Customers	3,900
Average Purchase Amount	\$59.76
Average Review Rating	3.75
Subscribers Share	27%

6. Business Recommendations

- **Boost subscriptions** — promote exclusive benefits to convert non-subscribers, who currently represent 73% of the customer base.
- **Strengthen customer loyalty programs** — reward repeat buyers to migrate them into the "Loyal" segment.
- **Review discount policy** — balance short-term sales lift against long-term margin impact, particularly for discount-dependent products such as Hats and Sneakers.
- **Optimize product positioning** — feature top-rated and best-selling products (e.g., Gloves, Jewelry, Blouse) more prominently in marketing campaigns.
- **Sharpen targeted marketing** — prioritize high-revenue age groups (Young Adult, Middle-aged) and Express-shipping users.